

Supplementary Evaluation and Statistical Analyses for “LLM-Assisted Reinforcement Learning for Affective Game Adaptation”

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This supplementary document accompanies the paper “LLM-Assisted Reinforcement Learning for Affective Game Adaptation,” which presents an affect-aware adaptive gaming architecture combining tabular reinforcement learning with bounded LLM-assisted exploration inside a MAPE-K loop. The system dynamically adapts gameplay and interface parameters in response to player affect and gameplay context, while preserving the transparency and inspectability of a discrete reinforcement learning controller.

This supplement provides the complete empirical and statistical material condensed from the main paper for space considerations. It includes full ANOVA and Friedman test tables, post-hoc Tukey and Wilcoxon comparisons, ANCOVA robustness checks, dominant-difficulty supplementary analyses, full post-hoc power analyses, questionnaire provenance details, additional descriptive statistics, and supplementary figures. These materials are provided to support transparency, reproducibility, and detailed inspection of the reported evaluation results.

CCS Concepts: • **Human-centered computing** → **User interface programming**; • **Software and its engineering** → **Software design engineering**; • **Computing methodologies** → **Reinforcement learning**; • **Applied computing** → **Computer games**.

Additional Key Words and Phrases: Adaptive games; Reinforcement learning; Large language models; Emotion; Personalization; Explainability.

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1 Overview

This document is organized to mirror the evaluation questions posed in the main paper “LLM-Assisted Reinforcement Learning for Affective Game Adaptation”. Section 2.1 describes the study design and methodology in detail. Sections 2.2 and 2.3 report the full statistical results for reward-relevant affect (RQ1) and event-level emotion effects (RQ2), including full ANOVA tables, ANCOVA robustness checks and all post-hoc Tukey and Wilcoxon tests. Section 2.4 contains the full results for the performance analyses (RQ3), including the dominant-difficulty supplementary analysis and Kruskal-Wallis robustness checks. Section 2.5 presents Q-table Convergence and Section 2.6 answers RQ4 about Performance-Emotion Relationship. The complete Friedman test outputs, questionnaire provenance table, and descriptive statistics for user-experience measures (RQ5) are presented in Section 2.7. Section 2.8 details the full power calculations and sensitivity analyses.

Author’s Contact Information: Blinded.

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2 Evaluation

We conducted a within-subjects experiment with 46 participants to evaluate the effects of LLM-assisted reinforcement learning on gameplay experience. Each participant experienced three counterbalanced conditions: No Adaptation (Baseline), Adaptive RL-only, and Adaptive LLM-assisted RL, followed by completion of questionnaires after each adaptation and a final collective questionnaire.

Our evaluation distinguishes between primary and secondary outcomes. Because the released controller is primarily driven by affective state, the primary outcomes of interest are affective and experiential. We nevertheless report progression-based gameplay measures as secondary corroborating signals, because adaptive-difficulty research often treats progression, challenge, and competence-related outcomes as complementary indicators of whether adaptations maintain a productive challenge-skill balance [1, 3, 6, 7].

2.1 Experiments Set-up

2.1.1 Participants. We recruited 46 individuals (22 female, 23 male, 1 non-binary) through university network channels. All participants reported prior gaming experience, though not necessarily with shooter games. Gaming experience levels ranged from non-gamers to hardcore gamers, with the majority identifying as casual or regular gamers.

2.1.2 Experimental Design. We employed a within-subjects design where each participant experienced all three adaptation conditions across three separate gaming sessions. To control for order effects, we used a Latin Square counterbalancing approach[2], which allows for efficient counterbalancing with fewer groups than a full factorial design while still controlling for first-order carryover effects. Participants were randomly assigned to one of three counterbalancing groups. Group 1 (participants 1-15) experienced the conditions in the order No Adaptation, Adaptive RL-only, then Adaptive LLM-assisted RL. Group 2 (participants 16-30) followed the order Adaptive RL-only, Adaptive LLM-assisted RL, then No Adaptation. Group 3 (participants 31-46) played Adaptive LLM-assisted RL first, followed by No Adaptation, then Adaptive RL-only. This design ensured that each condition appeared equally often in each position across the experiment.

Participants were not informed which adaptation condition they were playing during any session. The three rounds were simply labeled "Round 1-3," and participants were told that the game might behave differently across rounds. This blinding procedure allowed us to capture genuine perceptions of adaptation without biasing participant responses.

For participant-level condition comparisons, the within-subject design allows each participant to serve as their own control, thereby reducing between-subject variability. Event-level and pooled exploratory analyses are interpreted separately and with greater caution, as they do not explicitly control for participant-level fixed effects.

2.1.3 Apparatus. The experiment was conducted on a 27-inch display connected to a desktop computer running the Chrome web browser. Participants used an external keyboard for game controls and wore headphones for game audio. A 4K front-facing camera captured video for facial affect sensing. The adaptation engine ran locally within the web browser environment, with the LLM component accessed via API when required by the Adaptive LLM-assisted RL condition. To ensure independence across participants, the reinforcement learning agent was completely reset between participants.

2.1.4 Conditions. Three adaptation conditions were implemented. In the No Adaptation (Baseline) condition, the game ran with a fixed default configuration consisting of Regular difficulty, green visual theme, no horror sound effects, and no hint system. Game parameters remained constant throughout the session regardless of player behavior or detected affect. In the Adaptive RL-only

condition, a tabular Q-learning agent dynamically adjusted game parameters based on emotions and gameplay context. This condition made no LLM calls and relied entirely on the reinforcement learning policy. In the Adaptive LLM-assisted RL condition, the same Q-learning agent was employed, but a portion of exploration was replaced by LLM-guided exploration. The LLM provided context-aware action proposals during exploration phases, while learning and long-term policy optimization remained entirely within the RL agent. This hybrid approach allowed the system to leverage semantic understanding from the LLM while maintaining the learning capabilities of reinforcement learning.

2.1.5 Procedure. Each participant completed all three gaming sessions in a single visit. We ensured consistent initial gameplay conditions across all rounds and conditions. After completing each of the three gaming sessions, participants immediately filled out a per-session questionnaire. This questionnaire used 5-point Likert scales to assess six dimensions of their immediate experience: engagement (“How engaging was this gaming session?”), enjoyment (“How much did you enjoy this session?”), perceived difficulty (“How difficult did you find this session?”), awareness of adaptation (“Did you notice the game adapting to you during play?”), absorption (“I felt completely absorbed in the game”), and fairness (“The game did not feel like it was ‘cheating’”).

Once all three sessions were completed, participants filled out a final post-experiment questionnaire. This questionnaire gathered retrospective assessments across the entire experience. It included items about transparency (“I could understand why the game adapted to me”), satisfaction with progression (“I felt satisfied with the gameplay progression”), and a ranking task asking participants to rank the three sessions from most to least enjoyable. Additional items probed whether participants noticed differences between sessions, their understanding of how the adaptive system worked (“One or more sessions used AI to adapt the game to your emotional state. How much did you understand how the adaptation worked?”), their trust in the adaptive system, the importance they placed on knowing when a game is adapting, and their overall enjoyment of the entire experiment. All items used 5-point Likert scales except for the ranking task, which used ranks 1-3. Demographic information including gender and gaming experience level was collected at the end.

2.2 Emotion Arousal by Condition: Answer to RQ1

This analysis examined whether the three gameplay conditions differed in their arousal of the reward-relevant emotions used by the adaptive controller.

2.2.1 Process. The experiment used a within-subject design in which the same $N = 46$ participants completed all three conditions: No Adaptation (Baseline), Adaptive RL-only, and Adaptive LLM-assisted RL. Because the raw data consisted of time-sliced observations, emotion signals were first aggregated to the participant level within each condition. For each participant and condition, the mean arousal value was computed for Happy, Surprise, and Fear.

Following the reward structure used by the controller, we constructed an exploratory target-emotion score:

$$\text{TargetEmotion} = \text{Happy} + \text{Surprise} + \text{Fear}.$$

This composite is used as an omnibus indicator of reward-relevant affective activation, not as a validated unidimensional psychological construct. The three constituent emotions differ in valence and interpretation: Happy is positive affect, Surprise reflects arousal and novelty, and Fear may reflect desirable tension depending on context. Accordingly, the composite is useful for testing whether conditions differ overall on the affective dimensions used by the reward function.

Importantly, the composite should not be interpreted as a direct measure of player well-being, flow, or enjoyment.

A repeated-measures ANOVA was conducted with Condition as a three-level within-subject factor. When a significant main effect of Condition emerged, planned pairwise comparisons were used to assess differences between Baseline vs. RL-only, Baseline vs. LLM-assisted RL, and RL-only vs. LLM-assisted RL. Effect sizes are reported as partial η^2 .

2.2.2 Results. A repeated-measures ANOVA revealed a significant main effect of Condition on the combined TargetEmotion score, $F(2, 90) = 5.94$, $p = .04$, partial $\eta^2 = .117$ (Table 1). Mean target-emotion arousal was highest in the LLM-assisted RL condition relative to the other two conditions. Planned comparisons indicated that LLM-assisted RL significantly outperformed RL-only ($p = .021$). However, because three planned pairwise comparisons were conducted, this result is interpreted with multiplicity correction in mind. Under a Bonferroni-adjusted threshold of $\alpha = .017$, the LLM-assisted RL versus RL-only contrast does not provide strong corrected evidence of a reliable hybrid advantage on the composite metric. To verify that the effect of Condition on the TargetEmotion composite was not attributable to individual differences in player demographics, an analysis of covariance (ANCOVA) was conducted with Age and Experience as covariates. The effect of Condition remained significant, $F(2, 133) = 3.27$, $p = .041$, $\eta_p^2 = .047$, while neither Age, $F(1, 133) = 0.13$, $p = .722$, $\eta_p^2 = .001$, nor Experience, $F(1, 133) = 0.01$, $p = .904$, $\eta_p^2 = .001$, significantly contributed to the model. These findings confirm that the observed differences in reward-relevant affective activation across conditions were not confounded by participant demographics.

2.2.3 Discussion. The RQ1 result supports a cautious interpretation. The significant omnibus effect indicates that the three gameplay conditions differed in the affective signals used by the reward design. The LLM-assisted condition showed the strongest descriptive activation of the TargetEmotion composite. However, under Bonferroni correction for three pairwise comparisons, this contrast does not survive the adjusted threshold. We therefore treat the result as evidence that the conditions differ overall in reward-relevant affective activation, but avoid claiming that the hybrid controller robustly outperforms RL-only on TargetEmotion. This distinction is important because TargetEmotion aggregates affective dimensions with different theoretical meanings. The result is therefore best understood as an exploratory indicator that the adaptive mechanisms altered reward-relevant affective expression, rather than as proof that the LLM-assisted controller produced uniformly better emotional experience than RL-only adaptation.

Table 1. Repeated-Measures ANOVA Examining the Effect of Condition (Baseline, RL-only, LLM-assisted RL) on target emotions.

Tests of Within-Subjects Effects							
Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Condition	Sphericity Assumed	0.066	2	0.033	5.940	0.04	0.117

2.3 UI Parameters-Specific Emotion Effects: Answer to RQ2

This analysis examined whether specific UI/game adaptation parameters were associated with changes in emotional arousal during gameplay, and whether these relationships differed between the RL-only and LLM-assisted RL adaptive conditions.

2.3.1 Process. To examine whether specific UI/game adaptation parameters were associated with changes in emotional arousal, emotional change scores were computed at the level of individual adaptation events. Each gameplay timeslice represented approximately one second. Adaptation events were defined as transitions in which the *Adaptation name* changed from *No.adaptation* to an active adaptation parameter. For each adaptation event, emotional change was calculated using a ± 5 second window surrounding the event. The mean emotional intensity during the 5 seconds prior to adaptation (pre-window) was subtracted from the mean intensity during the 5 seconds following adaptation (post-window), producing change scores for happiness, fear, and surprise:

$$\Delta Emotion = \overline{Emotion}_{post(0-5s)} - \overline{Emotion}_{pre(0-5s)}$$

Separate two-way ANOVAs were conducted for each emotional change variable with *Adaptation Name* (9 levels) and *Adaptive Condition* (RL-only vs. LLM-assisted RL) as fixed factors. Tukey HSD post-hoc tests were performed for significant main effects of Adaptation Name. Partial eta squared (η_p^2) was reported as an estimate of effect size.

2.3.2 Results. For happiness change, the two-way ANOVA revealed (Table. 2) a significant main effect of Adaptation Name, $F(8, 1138) = 14.68, p < .001, \eta_p^2 = .094$, indicating that different adaptation parameters produced significantly different changes in happiness. A significant main effect of Adaptive Condition was also observed, $F(1, 1138) = 124.53, p < .001, \eta_p^2 = .099$, suggesting that the LLM-assisted RL system differed from the RL-only system in overall happiness change. In addition, the Adaptation Name $\times$ Condition interaction was significant, $F(8, 1138) = 2.53, p = .010, \eta_p^2 = .017$, indicating that the impact of specific adaptation types on happiness varied depending on the adaptive system employed.

For fear change, a significant main effect of Adaptation Name was found, $F(8, 1138) = 22.65, p < .001, \eta_p^2 = .137$, demonstrating that adaptation parameters substantially influenced fear responses. The main effect of Adaptive Condition was not significant, $F(1, 1138) = 0.07, p = .794$, indicating no overall difference in fear change between RL-only and LLM-assisted RL systems. The Adaptation Name $\times$ Condition interaction approached significance, $F(8, 1138) = 1.92, p = .054, \eta_p^2 = .013$, suggesting a potential but small system-dependent variation in fear responses.

For surprise change, a strong main effect of Adaptation Name was observed, $F(8, 1138) = 59.59, p < .001, \eta_p^2 = .295$, indicating that adaptation parameters strongly influenced surprise responses. The main effect of Adaptive Condition was not significant, $F(1, 1138) = 0.78, p = .378$. However, the Adaptation Name $\times$ Condition interaction was significant, $F(8, 1138) = 7.64, p < .001, \eta_p^2 = .051$, indicating that certain adaptation parameters elicited different levels of surprise depending on whether the RL-only or LLM-assisted RL system was used.

Post-hoc Tukey HSD comparisons further clarified which specific UI adaptations drove the observed emotional effects. For surprise, perceptually intense adaptations, particularly *Actions.Colorred* and *Actions.SoundHorror*, produced significantly greater increases compared to most other adaptation types (all $p < .001$). A similar pattern was observed for fear, where these same adaptations elicited significantly stronger increases relative to standard gameplay adjustments such as *Actions.Regular*, *Actions.Easy*, and *Action.PopupVisible*. These findings indicate that high-intensity visual and auditory modifications are especially effective in triggering arousal-based emotional responses.

In contrast, happiness followed an opposite pattern. Fear-related adaptations such as *Actions.Colorred* and *Actions.SoundHorror* were associated with significantly lower increases in happiness compared to more neutral or supportive adaptations, including *Actions.Regular*, *Actions.Easy*, and *Action.SoundNormal*. Together, these results suggest that sensory intensity manipulations primarily

amplify negative arousal (fear and surprise), whereas neutral or less fear-inducing adaptations are more conducive to maintaining or increasing positive affect.

Table 2. Two-way ANOVA results for emotional change as a function of Adaptation Type and Adaptive Condition

Two-Way ANOVA					
Emotion	Effect	df	F	p	ηp^2
Happiness	Adaptation Type	8, 1138	14.7	<.001	0.094
	Adaptation × Condition	8, 1138	2.53	0.01	0.017
Fear	Adaptation Type	8, 1138	22.7	<.001	0.137
	Adaptation × Condition	8, 1138	1.92	0.05	0.013
Surprise	Adaptation Type	8, 1138	59.6	<.001	0.295
	Adaptation × Condition	8, 1138	7.64	<.001	0.051

2.3.3 Discussion. The event-level analysis provides evidence that the concrete adaptation parameters implemented in CubeWars are associated with different short-term affective changes. Across happiness, fear, and surprise, Adaptation Name was a significant predictor of emotional change, indicating that the implemented levers were not affectively interchangeable. This supports the design premise that challenge regulation, sensory changes, and support prompts can shape player affect in different ways.

At the same time, these results should be interpreted as local evidence about the implemented CubeWars interventions rather than as general causal proof that these adaptation categories universally induce specific emotions. Since adaptation events occur repeatedly within participants, this event-level analysis is interpreted as exploratory and descriptive of short-term affective dynamics in the logged gameplay data. The results identify associations between adaptation events and subsequent emotion changes, but they do not replace participant-level condition comparisons. The effects are tied to the specific operational details of this artifact: red zombie color, horror background audio, wave difficulty changes, and modal popup support. In particular, the strong effect of Adaptation Name for surprise suggests that perceptual and environmental changes were especially salient to participants, while the fear results indicate that some sensory manipulations were associated with stronger arousal-related responses.

The condition-level effects were narrower. Adaptive Condition significantly influenced happiness but not fear or surprise, and interaction effects were significant for happiness and surprise. This suggests that the LLM-assisted controller did not simply increase all target emotions uniformly. Rather, the hybrid system appears to have changed how particular adaptation types were selected or experienced, especially for positive affect and surprise-related responses. This interpretation is more consistent with the architecture: the LLM does not replace the RL controller or directly optimize emotion; it biases a bounded portion of exploration toward contextually plausible actions that are then evaluated by the same reward mechanism.

The RQ2 results support the local face validity of the implemented adaptation levers and provide evidence that the emotional impact of adaptation depends on the specific parameter being changed. They also reinforce the need for caution: the results show differentiated short-term associations between adaptations and facial-expression outputs, but they do not establish universal mappings from game parameters to emotional states.

2.4 Adaptation vs. Baseline Performance: Answer to RQ3

This analysis examined whether experimental condition influences player performance. Performance was measured using two outcomes derived from gameplay logs: the maximum level reached and the maximum number of zombie kills achieved. Although maximum level reached and maximum zombie kills are correlated, they capture distinct aspects of gameplay performance. Kill count reflects short-term action throughput, including partial progress within levels, whereas level progression reflects sustained success across sequential waves. Reporting both metrics therefore provides complementary perspectives on performance. Performance is analyzed here as a secondary behavioral outcome rather than as the primary success criterion of the affective adaptive system. The online controller is designed around affective feedback, while performance measures provide behavioral triangulation: they indicate whether adaptation is accompanied by observable differences in progression. This distinction is important because adaptive difficulty can itself alter the opportunities for progression, for example by changing enemy spawn patterns or the difficulty of subsequent waves.

2.4.1 Process. To examine whether the experimental condition influenced player performance, a repeated-measures design was employed, as each participant experienced all three gameplay conditions: Baseline (No Adaptation), RL-only (MFRL-Adaptation), and LLM+RL (LLM-Adaptation). Performance was measured using two outcomes derived from gameplay logs: the maximum level reached and the maximum number of zombie kills achieved. For each participant and condition, the maximum level and maximum kill count were extracted and aggregated. A one-factor repeated-measures ANOVA was conducted for each performance outcome with Condition (three levels) as the within-subject factor. Multivariate statistics (Pillai’s Trace) were reported due to the robustness of this approach in repeated-measures designs. Pairwise comparisons were examined to determine specific differences between conditions.

2.4.2 Results. For maximum level reached, the repeated-measures ANOVA revealed a significant effect of Condition, as indicated by Pillai’s Trace, $F(2, 44) = 7.635$, $p = .001$, $\eta_p^2 = .258$. Pairwise comparisons showed that both adaptive conditions resulted in significantly higher maximum levels compared to the Baseline condition. Specifically, the LLM+RL condition yielded significantly higher levels than Baseline (Mean Difference = 1.283, $p = .003$), and the RL-only condition also significantly outperformed Baseline (Mean Difference = 1.848, $p < .001$). No significant difference was found between the RL-only and LLM+RL conditions ($p = .166$). Notably, the RL-only condition produced the largest numerical improvement in maximum level relative to baseline, although this advantage did not significantly exceed the performance observed under the LLM+RL system.

For maximum zombie kills, a significant effect of Condition was also observed, $F(2, 44) = 7.498$, $p = .002$, $\eta_p^2 = .254$ (Table.3). Pairwise comparisons indicated that both adaptive systems significantly increased kill counts relative to Baseline gameplay. The LLM+RL condition resulted in significantly more kills than Baseline (Mean Difference = 14.304, $p = .005$), and the RL-only condition produced a larger improvement over Baseline (Mean Difference = 22.000, $p < .001$).

To further verify whether the effect of Condition was attributable to individual differences in player demographics or not, an analysis of covariance (ANCOVA) was conducted with Age and Experience as covariates. The effect of Condition remained significant, $F(2, 133) = 6.89$, $p = .001$, $\eta_p^2 = .094$, while neither Age, $F(1, 133) = 0.34$, $p = .561$, $\eta_p^2 = .003$, nor Experience, $F(1, 133) = 0.10$, $p = .758$, $\eta_p^2 = .001$, significantly contributed to the model. These findings confirm that the observed performance differences across conditions were not confounded by participant demographics.

Supplementary Analysis: Controlling for Dominant Difficulty Exposure. To address the potential confound that adaptive conditions may have produced higher performance outcomes simply by

exposing participants to easier difficulty settings, a supplementary analysis was conducted. For each participant and adaptive condition, the dominant difficulty level was operationalised as the most frequently triggered difficulty adaptation event (Easy, Regular, or Hard). In the Baseline condition, the difficulty parameter was fixed at Regular throughout by design, no difficulty adaptations were triggered across all 15,375 timeslices (all recorded No. adaptation), confirming that all Baseline participants share the same difficulty setting. Dominant difficulty was therefore assigned as Regular for all Baseline participants.

In the adaptive conditions, difficulty exposure was distributed unevenly across participants: LLM+RL comprised Easy ($n = 18$), Regular ($n = 9$), and Hard ($n = 19$); RL-only comprised Easy ($n = 9$), Regular ($n = 19$), and Hard ($n = 18$). Importantly, this unequal distribution is not incidental, it reflects the fact that difficulty exposure in adaptive conditions is endogenous to the adaptation mechanism itself: the RL agent selected difficulty levels in direct response to each participant's real-time emotional and performance state. Controlling for difficulty statistically as an independent covariate would therefore risk removing part of the adaptation effect under investigation, since difficulty modulation is one of the primary channels through which adaptation operates. For this reason, the primary supplementary evidence is drawn from a conservative like-for-like comparison that avoids this confound.

Specifically, restricting the sample to participants whose dominant difficulty was Regular across all conditions ($n = 74$ observations: Baseline $n = 46$, LLM+RL $n = 9$, RL-only $n = 19$), a one-way ANOVA revealed that the condition effect on maximum zombie kills remained significant, $F(2, 71) = 7.62$, $p = .001$, and on maximum level reached, $F(2, 71) = 8.37$, $p < .001$. This indicates that performance advantages persisted even when all groups operated under equivalent difficulty exposure, with LLM+RL ($M = 85.89$, $SD = 28.46$) and RL-only ($M = 72.95$, $SD = 22.84$) both outperforming Baseline ($M = 58.39$, $SD = 19.61$). A non-parametric Kruskal-Wallis test confirmed these results, $H(2) = 12.73$, $p = .002$ for kills and $H(2) = 13.45$, $p = .001$ for level, ruling out sensitivity to distributional assumptions given the small cell sizes in the adaptive groups.

As an additional robustness check, dominant difficulty was included as an ordinal covariate (Easy = 1, Regular = 2, Hard = 3) in the full dataset. The effect of Condition remained significant for both maximum zombie kills, $F(2, 134) = 6.85$, $p = .001$, and maximum level reached, $F(2, 134) = 7.03$, $p = .001$, while dominant difficulty was not a significant predictor in either model (both $p > .10$). Collectively, these results confirm that the observed performance advantages of adaptive conditions over Baseline cannot be attributed to differential difficulty exposure, and that the condition effect reflects a genuine benefit of adaptation rather than a mechanical consequence of easier gameplay settings.

Table 3. Repeated-Measures ANOVA Results for Player Performance Across Experimental Conditions.

Multivariate Tests ^a							
Effect			Value	F	Hypothesis df	Error df	Partial Eta Squared
Max Level	Condition	Pillai's Trace	0.258	7.635 ^b	2.000	44.000	0.001
MaxZombies Killed	Condition	Pillai's trace	0.254	7.498 ^a	2.000	44.000	0.002

2.4.3 Discussion. The performance results show that adaptive gameplay improved absolute progression relative to the static baseline. Both RL-only and LLM-assisted RL led to higher maximum level reached and greater maximum zombie kills than the non-adaptive condition. This supports

the general value of runtime adaptation in CubeWars: when the game responds to the player’s state, participants progress further than when the game remains static.

However, the results do not show that LLM-assisted RL produces superior objective performance compared to RL-only adaptation. No statistically significant differences were observed between the two adaptive systems for either performance measure. The objective performance evidence supports an adaptive-versus-static advantage, but not a robust hybrid-versus-RL-only performance advantage.

These findings should also be interpreted with the mechanics of adaptive gameplay in mind. Maximum level reached and maximum kills are meaningful progression indicators, but they are not independent of game configuration. Because the adaptive controllers can alter difficulty and enemy spawning conditions, performance outcomes reflect both player behavior and system-driven changes to the challenge structure. For this reason, we treat performance as a corroborating behavioral signal rather than as the primary evaluation metric for an affect-aware adaptation system.

2.5 Q-table Convergence and Learning Dynamics

To examine whether the two adaptive systems developed stable learned strategies during gameplay, we analyzed logged action traces as a behavioral proxy for the underlying Q-table policies. Across both conditions, action selection became less variable over time, with later portions of sessions showing more repeated use of specific adaptations, indicating a shift from exploratory to more stable behavior.

Both conditions triggered adaptations at near-identical rates (LLM-assisted RL: 3.19% of timeslices; RL-only: 3.20%), with comparable per-participant counts ($M = 13.07$ vs. $M = 12.11$, $U = 1142.5$, $p = .510$). Action type distributions did not differ significantly between conditions, either at the level of individual actions, $\chi^2(8) = 5.10$, $p = .747$, or across broader categories (Difficulty, Audio, Support, Visual), $\chi^2(3) = 0.81$, $p = .848$. Both systems prioritised difficulty adjustments most frequently ($\approx 34\%$), followed by support, audio, and visual adaptations in similar proportions.

Taken together, these results suggest that both agents exhibited comparable within-session stabilization in high-level adaptation behavior. These findings should be interpreted as behavioral evidence of policy stabilization rather than proof of full theoretical Q-table convergence.

2.6 Performance-Emotion Relationship: Answer to RQ4

This analysis investigated whether target emotional engagement was associated with gameplay performance across experimental conditions.

2.6.1 Process. To examine whether emotional engagement was associated with gameplay performance, Pearson correlation analyses were conducted across all participant-condition observations ($N = 138$). Mean emotional engagement scores (MeanHappy, MeanFear, MeanSurprise) were computed for each participant and condition, alongside performance metrics (maximum level reached and maximum zombie kills). Two-tailed Pearson correlations were calculated to assess the relationship between emotional engagement and performance outcomes.

2.6.2 Results. A significant positive relationship was found between happiness and both performance measures. Mean happiness was moderately correlated with maximum level reached, $r = .473$, $p < .001$, and with maximum zombie kills, $r = .472$, $p < .001$. This indicates that higher levels of positive emotional engagement were associated with improved gameplay performance.

In contrast, fear was not significantly correlated with maximum level, $r = -.136$, $p = .111$, nor with maximum zombie kills, $r = -.140$, $p = .101$. Similarly, surprise was not significantly associated with maximum level, $r = -.043$, $p = .619$, or maximum zombie kills, $r = -.057$, $p = .506$.

2.6.3 Discussion. The correlation results provide triangulating evidence for the relevance of positive affect in adaptive gameplay. Sessions with higher average happiness tended to be associated with higher levels reached and more zombie kills. In contrast, fear and surprise were not significantly associated with either progression metric in the pooled session-level analysis. Given that each participant contributed observations in all three conditions, these correlations should be interpreted as exploratory session-level associations rather than as fully independent participant-level estimates.

These findings justify including performance as a secondary behavioral outcome. The paper does not treat performance as the primary optimization target of the affective controller; rather, performance provides a complementary signal indicating whether more positive affective sessions also correspond to more successful gameplay. The observed happiness-performance association is consistent with this interpretation.

However, the correlations should not be read causally. The analysis does not establish whether happiness improves performance, whether better performance makes players happier, or whether both are jointly influenced by adaptation and challenge level. The result is best interpreted as a session-level association supporting the broader triangulation of affective, behavioral, and subjective outcomes.

2.7 User Experience Evaluation: Answer to RQ5

2.7.1 Process. To address how participants perceived the three adaptation conditions in terms of engagement, satisfaction, explainability, and fairness, we analyzed responses from three condition-specific per-session questionnaires and one final post-experiment questionnaire administered during the study. After each of the three gaming sessions, participants completed a per-session questionnaire using 5-point Likert scales (1 = lowest, 5 = highest). This questionnaire assessed immediate perceptions of engagement ("How engaging was this gaming session?"), enjoyment ("How much did you enjoy this session?"), difficulty ("How difficult did you find this session?"), awareness of adaptation ("Did you notice the game adapting to you during play?"), absorption ("I felt completely absorbed in the game"), and fairness ("The game did not feel like it was 'cheating'").

Following completion of all three sessions, participants filled out a post-experiment questionnaire that gathered retrospective assessments. This included items measuring understanding of adaptation rationale ("I could understand why the game adapted to me"), satisfaction with progression ("I felt satisfied with the gameplay progression"), comparative enjoyment (ranking the three sessions from most to least enjoyable, where 1 = most enjoyable and 3 = least enjoyable), awareness of differences between sessions, understanding of how the adaptation worked, trust in the adaptive system, importance placed on knowing when adaptation occurs, and overall experiment enjoyment.

Given the within-subjects design where each participant experienced all three conditions, we employed non-parametric Friedman tests to assess differences across conditions for each metric. When the omnibus Friedman test indicated significant differences, we conducted post-hoc pairwise Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = .017$) to identify specific condition differences. For the post-experiment measures that were not condition-specific, we report descriptive statistics characterizing participants' overall perceptions of the adaptive system.

It is important to note that, since the user-study questionnaire uses adapted items rather than a validated full-scale deployment, we clearly distinguish between construct provenance and psychometric validation. The Game Experience Questionnaire (GEQ) remains widely used in games research, but later validation work found no evidence for the originally postulated seven-factor structure as commonly cited, and reporting practice around the instrument has often been inconsistent. The Player Experience of Need Satisfaction (PENS) derives from self-determination theory work on autonomy, competence, and relatedness in game experience, and later validation work

Table 4. Questionnaire provenance and interpretation in the revised manuscript.

Item	Construct traced to	Status	How to report it
How engaging was this gaming session?	Engagement-related experience	Adapted exploratory item	Report as condition-varying exploratory PX item
How much did you enjoy this session?	Enjoyment / positive experience	Adapted exploratory item	Report as condition-varying exploratory PX item
How difficult did you find this session?	Perceived challenge / difficulty	Adapted exploratory item	Report as condition-varying exploratory PX item
Did you notice the game adapting to you during play?	Adaptation awareness	Custom item	Report as condition-varying adaptation-awareness item
I felt completely absorbed in the game	Absorption / immersion-related experience	Adapted exploratory item	Report as condition-varying exploratory PX item
The game did not feel like it was “cheating”	Perceived fairness / acceptability	Custom item	Report as condition-varying acceptability item
I could understand why the game adapted to me	Perceived explainability	Custom item	Descriptive only (global perception)
I felt satisfied with the gameplay progression	Satisfaction / perceived competence	Custom item	Descriptive only (global perception)
Did you notice differences between the three sessions?	Manipulation salience	Custom item	Descriptive only (global perception)
How much did you understand how the adaptation worked?	Transparency understanding	Custom item	Descriptive only (global perception)
How much did you trust the adaptive system?	Trust	Custom item	Descriptive only (global perception)
How important is it to you to know when a game is adapting?	Transparency preference	Custom item	Descriptive only (global perception)
Overall, how enjoyable was the entire experiment?	Overall study enjoyment	Custom item	Descriptive only (global perception)

has treated both PENS and GEQ as instruments whose use benefits from explicit, item-level justification rather than causal scale borrowing [4, 5, 7, 8]. Several items were adapted from constructs commonly measured in such game-experience works, including engagement, challenge, and immersion/absorption, while other items (e.g., adaptation awareness, understanding, trust, and perceived fairness) were custom items introduced to capture adaptive-system transparency and acceptability. We therefore do not report GEQ or PENS subscale scores. Instead, we trace item provenance explicitly and report an internal-consistency estimate for an exploratory experience-and-responsiveness subset.

Table 4 maps each questionnaire item to its underlying conceptual construct and specifies whether it was adapted from prior game-experience work or introduced as a custom measure. This distinction is important because the questionnaire was designed as an exploratory adaptation-focused instrument informed by existing frameworks (e.g., GEQ and PENS), rather than as a direct deployment of validated GEQ or PENS scales. Accordingly, the table also indicates how each item is interpreted in the analysis, distinguishing condition-level comparisons from descriptive post-experiment items.

2.7.2 Results. Table 5 presents descriptive statistics and inferential test results for the six per-session subjective metrics. The analysis revealed substantial and consistent differences across conditions for four key metrics. Engagement ratings increased significantly across conditions, with the Baseline condition receiving the lowest ratings ($M = 2.96$, $SD = 0.82$), RL-only showing moderate engagement ($M = 3.59$, $SD = 0.72$), and LLM-assisted RL achieving the highest engagement ($M =$

Table 5. Descriptive statistics and Friedman test results for per-session subjective metrics across the three adaptation conditions. Values represent means with standard deviations in parentheses. All metrics used 5-point Likert scales.

Metric	Baseline	MFRL	LLM+MFRL	$\chi^2(2)$	p
Engagement	2.96 (0.82)	3.59 (0.72)	4.04 (0.59)	34.86	<.001
Enjoyment	2.91 (0.69)	3.46 (0.75)	3.93 (0.80)	26.17	<.001
Absorption	2.67 (0.67)	3.20 (0.93)	3.85 (0.84)	37.97	<.001
Noticed Adaptation	1.48 (0.59)	3.20 (0.88)	3.89 (0.80)	70.35	<.001
Fairness	3.65 (0.97)	3.61 (0.77)	3.89 (0.71)	5.82	.054
Difficulty	3.04 (0.89)	2.72 (0.69)	2.59 (0.86)	5.04	.080

4.04, SD = 0.59). The Friedman test confirmed these differences were significant, $\chi^2(2) = 34.86$, $p < .001$. Post-hoc tests revealed that all three pairwise comparisons were significant (all $p < .017$), indicating that each successive level of adaptation sophistication improved participant engagement.

Enjoyment ratings followed a similar pattern. Participants found the Baseline condition least enjoyable ($M = 2.91$, SD = 0.69), while RL-only offered moderate enjoyment ($M = 3.46$, SD = 0.75), and LLM-assisted RL was rated most enjoyable ($M = 3.93$, SD = 0.80). The omnibus test was significant, $\chi^2(2) = 26.17$, $p < .001$, and post-hoc analyses indicated that both adaptive conditions were rated more enjoyable than the static baseline. The LLM-assisted condition had the highest mean enjoyment rating.

The absorption metric, which assessed the extent to which participants felt immersed in the game, showed the most pronounced differences. Baseline sessions yielded low absorption scores ($M = 2.67$, SD = 0.67), RL-only produced moderate absorption ($M = 3.20$, SD = 0.93), and LLM-assisted RL generated strong absorption ($M = 3.85$, SD = 0.84), $\chi^2(2) = 37.97$, $p < .001$. All pairwise comparisons were significant (all $p < .017$), indicating that both forms of adaptation enhanced immersion, with the LLM-assisted approach offering additional benefits beyond standard reinforcement learning.

Perhaps most striking was the awareness of adaptation metric. Participants in the Baseline condition reported minimal perception of adaptive behavior ($M = 1.48$, SD = 0.59), which is consistent with the static nature of this condition. In contrast, both adaptive conditions were readily noticed, with RL-only rated at $M = 3.20$ (SD = 0.88) and LLM-assisted RL at $M = 3.89$ (SD = 0.80). The overall effect was highly significant, $\chi^2(2) = 70.35$, $p < .001$, and all pairwise comparisons differed significantly (all $p < .017$). Notably, participants were significantly more aware of the LLM-assisted adaptation compared to standard RL adaptation, suggesting that the LLM's contextual action proposals may have produced more noticeable or meaningful changes to gameplay.

In contrast to these strong effects, two metrics showed no significant differences across conditions. Fairness ratings were relatively high across all conditions (Baseline: $M = 3.65$, SD = 0.97; RL-only: $M = 3.61$, SD = 0.77; LLM-assisted RL: $M = 3.89$, SD = 0.71) and did not differ significantly, $\chi^2(2) = 5.82$, $p = .054$. This suggests that neither form of adaptation was perceived as unfair or as the game cheating. Similarly, perceived difficulty showed no significant differences across conditions, $\chi^2(2) = 5.04$, $p = .080$, though there was a non-significant trend toward both adaptive conditions being perceived as slightly easier than the Baseline.

Figure 1 shows the distribution of ratings across conditions for each metric, revealing consistently higher medians for LLM+MFRL and MFRL compared to Baseline in engagement, enjoyment, absorption, and noticed adaptation.

The ranking data provided additional evidence of participant preferences. When asked to rank the three sessions from most to least enjoyable, a clear hierarchy emerged. LLM-assisted RL was

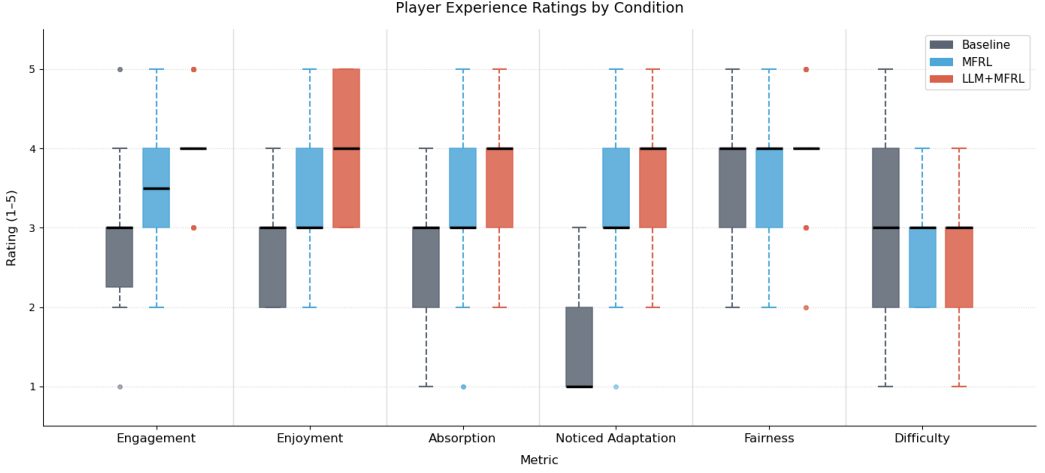


Fig. 1. Boxplots of player experience ratings (1–5) across six metrics for the three conditions (Baseline, MFRL, LLM+MFRL). The horizontal line inside each box represents the median and the box spans the interquartile range (IQR).

ranked as the most enjoyable session by 30 participants (65.2%), received a middle ranking from 16 participants (34.8%), and was never ranked as least enjoyable. RL-only was ranked first by 16 participants (34.8%), second by 28 participants (60.9%), and last by only 2 participants (4.3%). In stark contrast, the Baseline condition was ranked first by no participants, second by only 2 participants (4.3%), and last by 44 participants (95.7%). The mean rankings reflected this pattern (Baseline: $M = 2.96$, $SD = 0.21$; RL-only: $M = 1.70$, $SD = 0.55$; LLM-assisted RL: $M = 1.35$, $SD = 0.48$), with a highly significant Friedman test, $\chi^2(2) = 65.91$, $p < .001$.

The post-experiment questionnaire assessed participants’ overall perceptions of the adaptive system across all three sessions. Table 6 presents the results for seven items measured on 5-point Likert scales. Participants reported high awareness that the three sessions differed ($M = 3.98$, $SD = 0.65$), confirming that the manipulation was perceptible. Overall enjoyment of the experiment was relatively high ($M = 3.83$, $SD = 0.71$), as was satisfaction with gameplay progression ($M = 3.74$, $SD = 0.93$). Participants expressed moderate importance regarding knowing when a game is adapting ($M = 3.80$, $SD = 0.93$) and showed moderate trust in the adaptive system ($M = 3.59$, $SD = 0.72$). However, understanding of the adaptive mechanisms was more limited. While participants had some grasp of why the game adapted ($M = 3.35$, $SD = 0.92$), their understanding of how the adaptation worked was notably lower ($M = 2.98$, $SD = 0.95$), suggesting room for improved explainability in future implementations.

2.7.3 Discussion. The subjective results provide the clearest evidence for the experiential value of adaptation in CubeWars. Participants differentiated strongly between the three conditions, with adaptive conditions generally receiving higher ratings than the static baseline. The most consistent hybrid-specific advantages appeared in engagement, absorption, and awareness of adaptation, where the LLM-assisted condition showed the highest ratings.

These findings support a calibrated interpretation of the LLM’s contribution. The LLM-assisted controller should not be described as uniformly superior across all outcomes. Instead, its clearest added value lies in perceived responsiveness and experiential quality. Participants appeared more aware of adaptation in the LLM-assisted condition, and their engagement and absorption ratings

Table 6. Descriptive statistics for post-experiment questionnaire items assessing overall perceptions of the adaptive system. All items used 5-point Likert scales ($N = 46$).

Item	<i>M</i>	<i>SD</i>
I could understand why the game adapted to me	3.35	0.92
How much did you understand how the adaptation worked?	2.98	0.95
How much did you trust the adaptive system?	3.59	0.72
How important is it to you to know when a game is adapting?	3.80	0.93
I felt satisfied with the gameplay progression	3.74	0.93
Did you notice differences between the three sessions?	3.98	0.65
Overall, how enjoyable was the entire experiment?	3.83	0.71

were highest in that condition. This is consistent with the architectural role of the LLM as bounded exploration support: it may help the controller select adaptations that feel more contextually meaningful, while the RL mechanism remains responsible for learning from affective feedback.

At the same time, the subjective results are interpreted carefully. Enjoyment was highest descriptively in the LLM-assisted condition, but enjoyment should not be used alone as evidence that the hybrid controller robustly outperforms RL-only. Fairness and perceived difficulty did not differ significantly across conditions, suggesting that the adaptive mechanisms were not generally experienced as unfair or as making the game obviously easier or harder. This is encouraging for acceptability, but it also reinforces the need to separate perceived experience from objective performance.

The ranking task further supports the broad preference for adaptive gameplay over a static baseline. Participants overwhelmingly ranked the baseline as the least enjoyable condition, while both adaptive conditions were preferred. However, preference rankings should be interpreted together with the condition-specific ratings rather than as standalone proof of hybrid superiority. The strongest claim supported by the subjective data is that adaptive gameplay improved player experience relative to static play, and that LLM-assisted exploration showed its clearest benefit in perceived adaptivity, engagement, and absorption.

The post-experiment questionnaire also identifies a design challenge. Participants noticed differences between sessions and expressed moderate trust in the adaptive system, but their understanding of how the adaptation worked was more limited. This suggests that future systems should not only adapt effectively but also provide better explanations of when and why adaptation occurs. For engineering interactive computing systems, this points to an engineering implication: adaptive interactive systems that combine RL and LLM components should be designed for inspectability and user-understandable transparency, not only for closed-loop optimization.

Throughout the results, we distinguish carefully between statistically supported findings and descriptive trends. Where pairwise differences do not survive correction for multiple comparisons, we report them as exploratory rather than confirmatory. This ensures that all claims in the paper are directly grounded in the presented evidence.

2.8 Post-Hoc Power Analysis

Given the absence of prior studies comparing equivalent three-condition adaptive gameplay architectures, a formal a priori power calculation was not feasible at the design stage. Post-hoc power analyses were conducted to evaluate the adequacy of the achieved sample ($N = 46$) and to aid interpretation of non-significant findings, using achieved effect sizes at $\alpha = .05$.

Achieved power was estimated from the exact non-centrality parameter ($\lambda = F \times df_{\text{effect}}$) for each primary repeated-measures ANOVA. The RQ1 TargetEmotion analysis ($\eta_p^2 = .117$, Cohen’s $f = 0.36$) achieved power of .87. The RQ3 performance analyses yielded higher power: .93 for maximum level reached ($\eta_p^2 = .258$, Cohen’s $f = 0.59$) and .93 for maximum zombie kills ($\eta_p^2 = .254$, Cohen’s $f = 0.58$), consistent with their large effect sizes. The ANCOVA robustness check achieved power of .92 ($\eta_p^2 = .094$, Cohen’s $f = 0.32$). All RQ2 two-way ANOVAs achieved power of 1.00, attributable to the large event-level sample ($N = 1,158$ adaptation events).

A priori sensitivity analyses using the RM-ANOVA non-centrality formula, tested across plausible within-subject correlations ($\rho = 0.3\text{--}0.7$), indicated that 80% power would have required a minimum of $N = 9\text{--}19$ participants for the RQ1 effect and $N = 5\text{--}9$ for the RQ3 performance effects, confirming that the present sample of 46 was comfortably sufficient for the observed effect magnitudes. For the non-parametric Friedman tests in RQ5, Kendall’s W indicated medium-to-large effects for significant metrics (Engagement: $W = .38$; Enjoyment: $W = .28$; Absorption: $W = .41$; Noticed Adaptation: $W = .77$), and negligible effects for non-significant metrics (Fairness: $W = .06$; Difficulty: $W = .06$).

These results confirm that the significant findings reported across RQ1, RQ3, and RQ5 are well-supported by adequate statistical power. The null result for the RL-only versus LLM-assisted RL comparison on objective performance (RQ3) should nonetheless be interpreted cautiously: with $N = 46$, the study was powered to detect medium-to-large effects, but a small difference between the two adaptive systems cannot be ruled out and would require a larger sample to detect reliably.

3 Conclusion

This supplementary document provides the complete empirical and statistical materials accompanying the evaluation presented in the main ACM PACMHCI paper titled “LLM-Assisted Reinforcement Learning for Affective Game Adaptation”. It includes the full ANOVA and Friedman test results, post-hoc comparisons, robustness analyses, questionnaire provenance information, supplementary figures, and detailed power analyses underlying the reported findings. The main paper should be consulted for the primary discussion of the system architecture, adaptive framework, and condensed interpretation of the evaluation results. Together, the main manuscript and this supplement provide a complete account of the design, implementation, and evaluation of the proposed LLM-assisted reinforcement learning approach for affective game adaptation.

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